TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology
Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2023-24(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Design & Analysis of Algorithms

Time: 3 Hours **Max Marks:** 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33503	Course Outcome
CO1	To understand the foundational concepts, characteristics, and performance analysis of algorithms.
CO2	To explore algorithm design paradigms.
CO3	To classify classical algorithms and recurrence relations.
CO4	To develop efficient solutions using structured techniques.
CO5	To understand complexity classes and non-deterministic algorithms.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define recurrence relation.	CO1	2	2
b.	What is Greedy method?	CO2	2	2
c.	Define optimal substructure.	CO3	2	2
d.	What is Hamiltonian cycle?	CO4	3	2
e.	Define NP-Hard problem.	CO5	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain asymptotic notations with examples.	CO1	2	5
b.	Solve recurrence using Master Theorem.	CO1	2	5
c.	Explain algorithm analysis (best, worst, average case).	C01	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Merge Sort and analyze its complexity.	CO2	2	5
b.	Explain Quick Sort (randomized version).	CO2	3	5
c.	Explain Kruskal's and Prim's algorithm.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Dynamic Programming concept.	CO3	2	5
b.	Solve 0/1 Knapsack problem.	CO3	3	5
c.	Explain Chain Matrix Multiplication.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain N-Queen using backtracking.	CO4	2	5
b.	Explain Branch and Bound for TSP.	CO4	4	5
c.	Compare Backtracking and Branch & Bound.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain NP-Complete problems with examples.	CO5	2	5
b.	Discuss Cook's theorem.	CO5	4	5
c.	Explain complexity classes P, NP.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create